Retrieval-Augmented AI System for Reliable and Grounded Question Answering

Project Overview

A system that takes a user query, retrieves supporting passages from a document store, generates an answer grounded in those passages, and then checks the answer for claims that are not actually supported by the retrieved evidence—flagging likely hallucinations before the answer reaches the user. The workflow below breaks this into prompt-driven stages that can be demonstrated and defended stage-by-stage in a viva.

Step 1 — Query Understanding Prompt

Runs first, on the raw user question, to clarify intent and extract the retrieval target before touching the document store.
You will receive a user question. Do not answer it yet.
1. Restate the core factual claim(s) being asked about.
2. List 2–3 keyword phrases best suited for a document search.
Question: "{user_query}"
Output as: CLAIM: ... | SEARCH_TERMS: ...

Step 2 — Retrieval Prompt / Call

Uses the search terms from Step 1 to pull the top-k passages from the vector store or document index.
Retrieve the top {k} passages most relevant to: {search_terms}
Return each passage with its source ID and similarity score.
Do not summarize or alter passage text—return verbatim.

Step 3 — Grounded Generation Prompt

Generates a draft answer, but is explicitly restricted to the retrieved passages only.
Answer the user's question using ONLY the passages below.
If the passages do not contain enough information, say so—do not fill gaps from general knowledge.

<passages>
{retrieved_passages}
</passages>

Question: "{user_query}"
Answer, and after each sentence tag the passage ID it relies on, e.g. [P2].

Step 4 — Hallucination Detection Prompt

Cross-checks every sentence of the draft answer against the tagged source passage to catch unsupported claims.
For each sentence in the draft answer below, check whether its tagged source passage actually supports the claim made.

<draft_answer>
{draft_answer}
</draft_answer>

<passages>
{retrieved_passages}
</passages>

For each sentence output: SUPPORTED, PARTIALLY SUPPORTED, or UNSUPPORTED (HALLUCINATION) with a one-line justification.

Step 5 — Confidence Scoring & Flagging Prompt

Aggregates the per-sentence verdicts from Step 4 into a single answer-level confidence score and flag.
Given these per-sentence verdicts: {verdict_list}

Compute:
• Overall confidence score (0–100%) = supported sentences / total
• FLAG = "REVIEW NEEDED" if any sentence is UNSUPPORTED, else "OK"
Output: SCORE: {{score}}% | FLAG: {{flag}}

Step 6 — Final Output & Reporting

Not a model prompt—the display layer that presents the answer alongside its evidence and hallucination flags to the user/evaluator.
Render to the user:
1. The answer text, with unsupported sentences visually highlighted
2. Confidence score and flag from Step 5
3. Source passages used, with IDs, for one-click verification
Log query, draft answer, verdicts, and score for later evaluation (this becomes the dataset for measuring detector accuracy).

Flow Summary

Stage	Prompt/Step	Key Output
User question	Step 1 — Query Understanding	Claim + search terms
Document store	Step 2 — Retrieval	Top-k passages
Answer drafting	Step 3 — Grounded Generation	Tagged draft answer
Fact-checking	Step 4 — Hallucination Detection	Per-sentence verdicts
Scoring	Step 5 — Confidence Scoring	Score % + flag
Delivery	Step 6 — Output & Reporting	Answer + evidence + log

Core Principle

An answer only reaches the user after every claim has been traced back to a retrieved source—unsupported claims are surfaced, not hidden.

Related Topics

Retrieval-Augmented Generation (RAG) • Reliable and Grounded Question Answering • Large Language Models (LLMs) • Hallucination Detection • Evidence-Based Answer Generation • Semantic Search • Vector Databases • Prompt Engineering • Claim Verification • Fact Checking • Confidence Scoring • Source Attribution • Explainable AI (XAI) • RAG System Evaluation